

Yining Hua

Boston, MA

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Patient-safe AI tools for healthcare, clinical NLP, inferential and etiologic epidemiology.

Education

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|---|-------------------|
| • Harvard T.H. Chan School of Public Health , Epidemiology (Ph.D.) | Expected May 2028 |
| • Harvard Medical School , Biomedical Informatics (MSc.) | Sep 2021–Mar 2023 |
| • Smith College , Computer Science (B.A.) | Sep 2017–May 2021 |

Selected Research & Industry Experience

Squirrel AI

Seattle, WA (Remote)

PhD Research Intern — LLM/RAG Systems

June 2025 - Present

- Designed and implemented an event-centric, graph-based memory-augmented journaling agent (entity and relation extraction with embedding-based indexing; retrieve, re-rank, and explain pipeline). Achieved **96-98%** inference cost reduction and **38%** lower latency versus sending all memories per turn to the base model, saving **\$0.143 per turn** on average (approximately **\$77 per user per 3 months**) while improving relevant-memory recall by **24%**.
- Develop an ESSA-aligned evaluation strategy for an adaptive math program in YMCA after-school settings, defining research questions, study design, implementation and training plan, standardized outcome measurement, fidelity and dosage monitoring, and a prespecified analysis plan with subgroup and dose-response components.

Department of Digital Psychiatry, Beth Israel Deaconess Medical Center

Boston, MA

Research Scientist — LLM Evaluation and Regulation in Healthcare

2024 – Present

- Led review-based research characterizing the mental health LLM landscape, resulting in publications in **Nature Portfolio** journals and **World Psychiatry** (IF=80).
- Contributing author, 2nd edition of *Artificial Intelligence in Behavioral and Mental Health Care* (ed. David D. Luxton), chapter on AI for mental health research.

Department of Neurology, Massachusetts General Hospital

Boston, MA

PhD Researcher — ALS Prognostics, LLMs & Digital Twins

October 2025 – Present

- Develop risk prediction models for survival, hospitalization, and initiation of noninvasive ventilation or PEG using landmarking and time-varying Cox models, sequence models, calibration auditing, etc.
- Develop digital twin models (synthetic counterfactuals) integrating EHR and trial data to simulate patient trajectories and counterfactual outcomes; apply synthetic control arms and policy evaluation under real-world covariate shift.

Dandelion Health

Boston, MA

PhD Intern — Core Analyst & Study Designer

July 2025 – Present

- Led a **\$650K** real-world evidence collaboration on vaccine adverse events with a top-10 pharmacy partner; delivered protocol and analysis in approximately **2 months** without rework.
- Target trial emulation of GLP-1 therapies: processed over **1.1M** echocardiography records and assembled a risk set of **373K** patients; developed an approximation algorithm that reduced cohort build and run time from **35 hours to 0.4 hours** per cycle (**~99%**).

Harvard Medical School and Affiliated Hospitals

Boston, MA

Research Assistant — Clinical NLP/LLMs, Social Media Epi, Data Platforms

2021 – 2025

- Boston Children's Hospital (Urology)**: Developed LLM-based extraction of multi-class outcomes from urodynamics charts; evaluated active learning strategies with clinician-in-the-loop review and calibration and error auditing. Trained BioClinicalBERT, LoRA-adapted Llama 3.1 8B-Instruct, and prompt-tuned GPT-4o (0-shot and 3-shot), improving clinician labeling accuracy by **6%** across two retraining and error-analysis cycles.

2. **Brigham & Women's Hospital (Internal Medicine)**: (1) Harmonized a Twitter corpus of **396,380,339** COVID-19 posts by integrating six datasets; resource supported **10+** publications. Designed studies of drug misinformation and implemented a Sentence-BERT with k-means pipeline to surface motivational themes; work cited by the **Agency for Healthcare Research and Quality**. (2) Developed multi-modal prediction models combining structured EHR data and clinical notes to identify patient subgroups; fine-tuned RoBERTa-based models and built a reusable training and preprocessing pipeline supporting **5+** subsequent lab studies and **3** top-journal publications.
3. **Harvard Medical School (CELEHS)**: Constructed a cross-ontology clinical dataset of **890,000+** concept, synonym, and relation triplets via web-based collection and harmonization across ICD, SNOMED CT, and RxNorm; improved mapping accuracy by **>30%** versus exact matching and enabled downstream ML studies.

Selected Publications & Preprints (denotes equal contribution)*

1. **Hua Y**, Na H, Li Z, Liu F, Fang X, Clifton D, Torous J. *A scoping review of large language models for generative tasks in mental health care*. **npj Digital Medicine** 8:230 (2025).
2. **Hua Y**, Siddals S, Ma Z, Galatzer-Levy I, Xia W, Hau C, Na H, Flathers M, Linardon J, Ayubcha C, Torous J. *Charting the evolution of AI mental health chatbots: from rule-based to large language models*. **World Psychiatry** 24:383-394 (2025).
3. **Hua Y**, Xia W, Bates D.W., Hartstein L., ..., Torous J. *Standardizing and scaffolding healthcare AI-chatbot evaluation*. **JMIR AI**. Accepted (2025).
4. Liu F, Li Z, Zhou H, ..., **Hua Y**, ..., Clifton D.A. *Large language models are poor clinical decision-makers: a comprehensive benchmark*. In: **EMNLP** (2024).
5. Liu F, Zhou H., Gu B., ..., **Hua Y**, ..., Clifton D.A. *Application of large language models in medicine*. **Nat Rev Bioeng**. 1-20 (2025).
6. Liu J*, Zhou P*, **Hua Y***, ... *Benchmarking Large Language Models on CMExam—A Comprehensive Chinese Medical Exam Dataset*. **NeurIPS** (2023).
7. Dwyer B., Flathers M., Sano A., ..., **Hua Y**, ... *MindBenchAI: an actionable platform to evaluate LLMs in mental healthcare*. **NPP Digit Psychiatry Neurosci**. Accepted (2025).
8. Ma J., Na H., Wang Z., **Hua Y**, Liu Y., Wang W., Chen L. *Detecting Conversational Mental Manipulation with Intent-Aware Prompting*. **COLING 2025**, pp. 9176-9183.
9. **Hua Y**, Beam A, Chibnik L.B., Torous J. *From statistics to deep learning: using large language models in psychiatric research*. **Int J Methods Psychiatr Res**. 34(1):e70007 (2025).
10. **Hua Y**, Blackley S., Shinn A., Skinner J., Moran L., Zhou L. *Identifying psychosis episodes in psychiatric admission notes via rules, ML, and pre-trained LMs*. **Translational Psychiatry**. Accepted (2025).
11. **Hua Y**, Wang L., Nguyen V., Rieu-Werden M., McDowell A., Bates D.W., Zhou L. *A deep learning approach for transgender and gender diverse patient identification in EHRs*. **J Biomed Inform**. 147:104507 (2023).
12. Wang Z., Na H., Gao R., Ma J., **Hua Y**, Chen L., Wang W. *From posts to timelines: modeling mental health dynamics from social media timelines with hybrid LLMs*. **CLPsych 2025**.
13. **Hua Y**, Jiang H., Lin S., Yang J., Plasek J.M., Bates D.W., Zhou L. *Using Twitter to understand perceptions of approved vs. off-label COVID-19 medications*. **JAMIA**. 29(10):1668-1678 (2022).
14. Ye Q., Liu J., Chong D., Zhou P., **Hua Y**, ... *Qilin-med: multi-stage knowledge injection medical LLM*. arXiv:2310.09089 (2023).
15. Liu J*, Wang Z*, Ye Q*, Chong D*, Zhou P*, **Hua Y***. *Qilin-Med-VL: towards a Chinese vision-language model for healthcare*. arXiv:2310.17956 (2023).
16. **Hua Y**, Lin S., Li M., Zhang Y., Foer D., Zhou P., Zhou L., Yang J. *Streamlining social media information retrieval for COVID-19 research with deep learning*. **JAMIA**. 31(7):1569-1577 (2024).
17. Jiang H*, **Hua Y***, Beeferman D., Roy D. *Annotating the Tweebank corpus on named entity recognition and building NLP models for social media analysis*. In: Proceedings of the 13th **Language Resources and Evaluation Conference (LREC)**, 2022.

Awards, Fellowships & Funding

- **COLING 2025**: Best Short Paper.
- **NeurIPS 2022**: Spotlight (long paper); **NeurIPS 2023**: Travel Award.

- **IEEE ICHI 2023:** Best Poster; **AMIA 2022:** Distinguished Poster.
- **Harvard T.H. Chan:** Rose Traveling Fellowship (2024); Brian & Heidi MacMahon Epidemiology Educational Fund (2023-2024).
- **Sigma Xi 2021:** Outstanding Student Researcher nominee.

Review Service

- **Nature Portfolio:** *Nature Medicine* (2024-2025), *Nature Mental Health* (2025), *npj Digital Medicine* (2024-2026).
- **JMIR Family (digital health):** *JMIR*, *JMIR Mental Health*, *JMIR Public Health and Surveillance* (2024-2025).
- **Medical Informatics Journals and conferences:** *JAMIA* (2023-2024), *Int. J. Medical Informatics* (2022, 2024), *AMIA Annual Symposium* (2022-2023), *AMIA Informatics Summit* (2023).
- **ML/AI Venues:** *ACL* (2025), *NeurIPS* (2023-2024), *ML4H* (2023), *COLING* (2022).

Teaching

- EPI 236: Analytical Clinical Epidemiology; Harvard T.H. Chan (Summer 2026)
- EPI 288: Intro to ML & Risk Prediction; Harvard T.H. Chan (Spring 2025 & Spring 2026)
- PHS 2000A: Quantitative Research Methods; Harvard T.H. Chan (Fall 2024)
- BMI 707 / EPI 290: Deep Learning for Biomedical Data; Harvard Medical School (Spring 2023)
- S-043 / Stat 151: Multilevel & Longitudinal Models; Harvard GSAS (Summer 2021)
- Data Science in Action: Machine Learning for Self-Driving Cars; Harvard T.H. Chan (Summer 2020)
- PHY 215: Light, Relativity, and Quantum Physics; Smith College (Spring 2020)
- PHY 210: Math Methods of Physical Sciences and Engineering; Smith College (Fall 2019)